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IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R26-IT-050

1. Topic (12 words max)

AI-Powered Unified Sri Lankan Tourism Platform Using GraphRAG and LLM Orchestration

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Information System Engineering(ISE)

4. If a continuation of a previous project:

Project ID	R26-IT-050
Year	2026

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Research Problem

Nature and Gravity of the Problem Sri Lanka tourism industry is now recovering fast, and arrivals are expected to hit nearly 3 million by end of 2025. But, the main problem is that the tourism ecosystem is very fragmented. Right now, travelers have to use many different apps for hotels, transport, and finding guides. There is no single "unified" platform. This is a big issue for the **User Persona**: the *Independent International Traveler*. These tourists want everything in one place, but they struggle with language barriers, fake reviews, and not knowing if a driver or guide is officially registered.

Impact of the Problem

Social & Economic: Because planning is so hard, many tourists only visit popular places like Ella or Galle. This means rural areas don't get enough money from tourism. Also, travelers spend too much time on logistics instead of enjoying the trip.

Technical: Most current apps just use simple search filters. They cannot handle complex requests like "Find a luxury hotel that is also near a yoga center and a vegan restaurant" in one go. There is a lack of "intelligent reasoning" in current Sri Lankan travel apps.
Safety & Environment: Sri Lanka has unique risks like flash floods or elephants crossing roads. Global apps like Google Maps don't show these local hazards in real-time. This puts foreign tourists in danger because they don't have "live" safety alerts.

Research Cluster & Industry Verticals

Research Cluster: AI & Machine Learning / Software Engineering.
Industry Vertical: TravelTech and Public Safety.

How the Problem attains SDGs

- **SDG 8 (Decent Work):** The platform helps local guides and drivers to find more work by connecting them directly to tourists.
- **SDG 9 (Innovation):** Using GraphRAG and AI Orchestration is a new way to build digital infrastructure in Sri Lanka.
- **SDG 11 (Sustainable Communities):** Real-time hazard mapping makes travel safer for both the visitors and the local people.

References,

[1] Sri Lanka Tourism Development Authority, "Year in Review 2024: Annual Statistical Report," SLTDA, Colombo, Sri Lanka, Rep. Jan. 2025. [Online]. Available: https://www.sltda.gov.lk/

[2] United Nations Development Programme (UNDP) Sri Lanka, "Digital Public Infrastructure (DPI) Roadmap: Transforming Tourism and Local Administration," UNDP Sri Lanka, Colombo, Sri Lanka, Rep. 2024. [Online]. Available: https://www.undp.org/srilanka

[3] D. Edge et al., "From Local to Global: A GraphRAG Approach to Query-Focused Summarization," Microsoft Research, Redmond, WA, USA, Tech. Rep. arXiv:2404.16130, Apr. 2024. [Online]. Available: https://arxiv.org/abs/2404.16130

[4] S. Wijesundara and T. Kumara, "Digital Transformation and the Evolution of the Smart Tourism Ecosystem in Sri Lanka: A Post-Pandemic Analysis," International Journal of Sustainable Tourism (IJST), vol. 1, no. 2, pp. 45-62, July 2025. [Online]. Available: https://arts.cmb.ac.lk/ijst

[5] X. Zhang, L. Wang, and J. Chen, "TravelRAG: A Tourist Attraction Retrieval Framework Based on Multi-Layer Knowledge Graph," ISPRS International Journal of Geo-Information, vol. 13, no. 11, p. 414, Nov. 2024. doi: 10.3390/ijgi13110414.

[6] R. S. Perera, "Tourism Chatbot Framework: Enhancing Visitor Experience Through GraphRAG and AI Chatbot," in Proc. Int. Conf. Adv. ICT for Emerging Regions (ICTer), Colombo, Sri Lanka, 2024, pp. 112-119.

6. Brief description of Existing Research and Systems (300 words max)

Summary of Existing Products/Systems Current tourists in Sri Lanka mostly use a mix of global and local apps. Global platforms like **Booking.com** or **TripAdvisor** are good for finding hotels, but they don't have good details about local transport or certified guides. For travel, people use **PickMe** or **Uber**, but these apps only work well in cities like Colombo and don't help with planning a whole tour. The **SLTDA (2024)** has some websites for information, but they are mostly static and not interactive. There is no single system that connects every part of the trip—accommodation, drivers, and safety alerts—into one "unified" app.

Summary of Existing Research Recently, many researchers are looking at **Large Language Models (LLMs)** for travel. A framework called **TravelRAG (2024)** showed how Knowledge Graphs can make attraction suggestions more accurate. Also, some local studies by **Bandara (2025)** mentions that Sri Lanka needs better "context-aware" apps to help with safety at places like Sigiriya. Globally, there is a lot of talk about **GraphRAG (2024)**. This is a new AI method that helps computers understand complex relationships, like matching a user's food preference with a specific hotel's kitchen history.

Gaps and Limitations Identified

- **Fragmented Apps:** Travelers have to switch between 4 or 5 different apps to manage one single trip, which is very annoying and confusing.
- **No Real-time Safety:** Existing maps like Google Maps don't know about Sri Lankan problems like **Human-Elephant Conflict** or sudden landslides during monsoon.
- **Fake Reviews:** Most current platforms have "star ratings" that can be faked. There is no AI system to check if a review is actually trustworthy or not.
- **Emergency Gap:** If a tourist has an accident, there is no direct digital link to services like **1990 (Suwa Seriya)** that can send their location automatically.
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References,

[1] Sri Lanka Tourism Development Authority (SLTDA), "Annual Performance Report," SLTDA, Colombo, Sri Lanka, 2024.

[2] Microsoft Research, "From Local to Global: A GraphRAG Approach to Query-Focused Summarization," Microsoft Research, Redmond, WA, USA, 2024.

[3] P. M. Bandara, "Digital Solutions for Safe Tourism in Sri Lanka," Academic Study, 2025.

[4] TravelRAG, "Tourist Attraction Retrieval via Multi-Layer Knowledge Graphs," *ResearchGate*, 2024.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

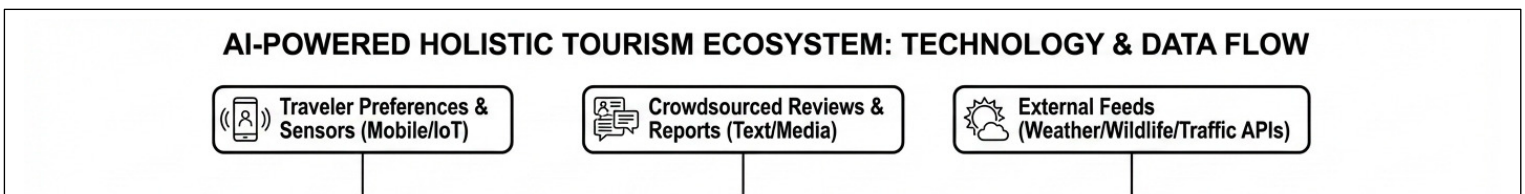
The proposed AI-Driven Tourism Intelligence Framework is an integrated smart-tourism ecosystem designed to deliver safer, smarter, and personalized travel experiences. It combines Graph-based Retrieval Augmented Generation (GraphRAG), community-driven cognitive nudging, incident-aware contextual mapping, and an autonomous guardian mobility grid into a unified AI platform. The solution bridges personalization, trust, safety, real-time awareness, and emergency response, areas where conventional tourism platforms fail.

At the core, the platform aggregates structured tourism assets such as hotels, attractions, transportation, activities, and guides into a Knowledge Graph. Using GraphRAG, the system traverses relationships within this graph to generate multi-criteria itineraries, satisfying combined traveler preferences like budget, dietary needs, wellness interests, accommodation standards, and transport preferences. Unlike traditional travel systems that treat selections independently, this framework creates cohesive, constraint-aware itineraries.

Alongside personalization, the platform introduces a Community-Driven Cognitive Travel Nudging System (CCTNS). Travelers share real-time reviews, safety tips, hygiene feedback, and service quality updates. AI filters out unreliable information and calculates dynamic Trust Scores instead of using static star ratings. Travelers get timely suggestions like safety alerts, better-value options, or service quality warnings, helping them make more confident and trustworthy decisions.

To improve safety, the framework uses Incident-Aware Contextual Mapping powered by AI. It combines crowdsourced incident reports with data from wildlife tracking, weather, and traffic sources. The AI checks each report for credibility, severity, and location, then marks real-time hazard zones like elephant crossings, flood-prone areas, or unrest zones. It also suggests safer routes. This turns regular tourism maps into dynamic safety tools.

The Autonomous Guardian & Mobility Grid helps people stay safe during emergencies. Its Zero-Latency Emergency Bridge connects directly with national emergency services, sending location and health data right away and overcoming language differences. If a vehicle breaks down, the Rescue Ride module quickly sends trusted support vehicles nearby to keep travel going. AI-powered triage sorts medical, environmental, and mechanical emergencies so the right help arrives fast.



8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

Gaining Specialized Domain Expertise To understand the Sri Lankan tourism sector, our team will use several official resources. First, we will study the **Sri Lanka Tourism Development Authority (SLTDA)** annual reports to understand tourist arrivals and behavior. Some members will also refer to the **Sri Lanka Institute of Tourism and Hotel Management (SLITHM)** website and guidelines to understand the official standards for registered guides and hotels. Also, we plan to consult with a few local "Good Samaritan" drivers to understand the real problems they face on the road, especially regarding vehicle breakdowns and wildlife hazards like elephant crossings.

Technical Knowledge and Skills Since our project uses advanced AI like **GraphRAG** and **LLM Orchestration**, we need to learn specific technical skills:

- **Member 1 (Itinerary):** Will follow the **Microsoft Research GraphRAG** documentation and open-source tutorials to learn how to build Knowledge Graphs from unstructured text.
- **Member 2 & 3 (Trust & Safety):** Will focus on **Natural Language Processing (NLP)** for sentiment analysis and **API Integration** (OpenWeather, Google Maps).
- **Member 4 (Guardian):** Needs to learn about **Real-time Data Streaming** and how to integrate mobile sensors (accelerometer) for crash detection, while also studying the digital dispatch protocols of the **1990 Suwa Seriya**ambulance service. All members will use online platforms like Coursera and YouTube for learning **Python** and **Gemini API** implementation.

Data Collection Strategy This project mainly uses an **API-First Approach**, meaning we don't need a massive manual database.

- **Official Data:** Information about hotels and certified guides will be gathered from the **SLTDA "Registered Service Providers"** database.
- **Environmental Data:** Real-time weather and disaster alerts will be fetched from the **Disaster Management Centre (DMC)** and the **Department of Meteorology** public feeds.
- **Map Data:** We will use **Google Places API** to get locations, opening hours, and traffic details.
- **Crowdsourced Data:** The "Event Fusion" and "CCTNS" modules will collect live reports (text/photos) directly from users.

Ethical Clearance Yes, this project requires **Ethical Clearance** for two main reasons:

- **Human Participants:** We are collecting crowdsourced safety reports and user reviews. We must ensure that user data is anonymized and that we have "Informed Consent" before storing any location data.
- **Emergency Services:** Our "Guardian" module interacts with emergency APIs. We must ensure our testing doesn't send "fake" alerts to real services like **1990** or **119**. We will follow the **University Ethics Review Committee (ERC)** guidelines to handle this safely.

Referees,

[1] Sri Lanka Tourism Development Authority (SLTDA), "Monthly Tourist Arrivals Reports," SLTDA, Colombo, Sri Lanka, 2025.

[2] Disaster Management Centre (DMC), "Early Warning and Emergency Operation Procedures," Ministry of Defence, Sri Lanka, 2025.

[3] Microsoft Research, "From Local to Global: A GraphRAG Approach to Query-Focused Summarization," Microsoft Research, Redmond, WA, USA, 2024.

[4] 1990 Suwa Seriya Foundation, "Technology in Pre-hospital Care: AI and Mixed Reality in Sri Lanka," 1990 Suwa Seriya Foundation, Sri Lanka, 2025.

9. Objectives and Novelty

Main Objective

To design and develop integrated AI-orchestrated tourism ecosystem that leveraging Graph Retrieval-Augmented Generation (GraphRAG), Large Language Models (LLM), Natural Language Processing (NLP), and Multi-Modal AI Event Fusion to automate and enhance end-to-end travel experience for Sri Lanka tourism including personalized multi-criteria itinerary generation, dynamic community-driven trust modeling for service validation, real-time incident-aware safety mapping using crowdsourced data fusion, and autonomous emergency response bridging, enabling seamless trip planning, context-aware risk mitigation and intelligent, language-agnostic tourism orchestration.

Member Name with Registration No	Sub Objective	Tasks	Novelty
Wijesinghe I.D.W. IT22555458	Design and implement a dynamic AI-driven Safety Intelligence Layer that utilizes Multimodal Event Fusion (text, image, and voice) and LLM-based reasoning to transform static maps into real-time "living" hazard	• Develop an AI Event Fusion Engine that merging unstructured, crowdsourced user reports (captured via mobile inputs) with structured external data from Weather APIs, Disaster Management Center (DMC) alerts, and Wildlife Movement feeds to creating a unified safety data stream. • Implement LLM-driven Semantic Analysis using Gemini for interpret the intent, severity, and urgency of	This component introduces a context-aware safety ecosystem that replace static tourist maps with a self-evolving hazard intelligence layer. Unlike traditional navigation apps which focus primarily on traffic, this research uniquely fuses LLM-based semantic reasoning with localized Sri Lankan risk factors (such as Human-Elephant Conflict and monsoon-driven landslides). It provides a "Living Map" where every tourists acts as a sensor,

	dashboards, enabling context-aware risk prediction and autonomous safety rerouting for tourists in Sri Lanka.	natural language reports (e.g., "The river is overflowing at the bridge") to automatically categorize incidents without any human moderations. • Engineer a Trust-Architecture & Cross-Validation System that assign reliability scores to user-reported hazards by cross-referencing them with official sensor data and nearby user-validations for prevent false alerts or "spam" reports. • Create a Contextual Mapping Interface that visualize these hazards as dynamic "Heat Zones" or "Red Segments" and push real-time safety protocols (e.g., "Elephant Corridor Active: Stay inside vehicle") based on the traveler's precise geolocations.	verified by AI against official APIs, ensuring that travel guidance is not just data-informed but contextually aware and real-time responsive, addressing the fatal limitations in static data for unpredictable tropical environments.
Madusha.M. IT22214720	Developing and executing an AI-based Autonomous Guardian & Mobility Grid that facilitates instant emergency response, smart crisis triaging, and	• Design an Emergency Integration Layer with Zero Latency to enable a direct API link through AI management for national emergency numbers (119 Police/1990 Suwa Seriya). • Establish and implement an electronic emergency data packet	Introduces the concept of a Zero Latency AI Emergency Bridge that can circumvent the use of voice channels for emergency calls to avoid language issues that cause delay in responding to emergencies. Suggests a two-layer safety and mobility framework that is a blend of official

	uninterruptible touristic mobility for visitors, via the integration of zero-latency emergency services, sensor-driven crisis detection, and crowd-driven rescue solutions in the Sri Lankan touristry scenario.	transmission system that can instantaneously deliver real-time geolocation data, identifiers, and medical information. • Design an activation system for Panic Mode with a crash detection mechanism using an accelerometer sensor in order to begin self-operated safety processes. • Develop an AI-assisted module for Contextual Crisis Triage to differentiate between a medical emergency and a mechanical failure using sensor information and usage patterns. • Develop the Crowdsourced Mobility Recovery algorithm ("Rescue Rides") that would allow it to find nearby verified vehicles within a specified radius and organize peer mobility recovery assistance. • Create a broadcast and matching system for informing nearby genuine users and finding a	rescue efforts and community-rescue rides. Employs AI crisis triaging with awareness of sensors, making it possible to correctly distinguish medical crises from mechanical failures without human interaction. Unveils Crowdsourced Mobility Continuity, a feature that ensures tourist routes are completed even in cases of vehicle breakdowns, instead of ending prematurely. Offers a tourism-focused resilience model, customized for Sri Lanka, turning the management of emergencies into a proactive, intelligence-driven safety environment.

		suitable rescue ride for stranded travelers in real time. • It must be tested and proven in a simulated accident, medical, or break-down situation to verify its response accuracy, latency, and system reliability.	
Deniyawatta D.G.A.D.N IT22895196	Develop an intelligent tour package generator that creates customized complete levels of itineraries based on specific user preferences. Represent tourism based data that analyze and structured through knowledge graph And applies graph based reasoning to match the multiple user preferences and integrates real time	• Collect and define sample tourism data such as hotels, drivers, vehicles and activities and identify key attributes such as location, service type, availability, that are required for itinerary generation. • Develops a knowledge graph to represent tourism data and their relationships. Nodes represent the data (Entities) and edges represent the relationships such as availability, location. • Converting user user travel requirements into structured constraints. Ex: Converting user inputs such as preferred location, accommodation type, food preferences type etc… are	• Propose the graph based augmented generation (GraphRAG) to generate a complete multi criteria tour packages while traditional systems only suggest separate level of components • Ensure the itinerary coherence by reasoning over tourism entities by allowing hotels, transport, activities etc… to be selected as one integrated package rather than independent items. • Performing constraint-satisfaction-based itinerary generation where all user requirements are validating before finalising the final

	data including travel time, traffic conditions, weather and opening hours to ensure the generated itineraries are practical and feasible.	translated into machine readable conditions. • Perform graph traversal and filtering to identify user specific and compatible tour components. The system can navigate through the knowledge graph and find combinations of hotels, activities, and drivers etc… to satisfy all the user preferences. • Integrating external APIs to optimize travel routes, travel times,and schedules to ensure whether these generated itineraries are timely and practically feasible or not. • Use a LLM to generate clear and natural language descriptions based strictly on the graph selected results.	package and reducing impractical and incomplete itineraries. • Grounds large language model (LLM) outputs in graph selected data and improves the reliability and significantly minimises the hallucinations of generated itineraries. • Checking whether it is practically and executable in real world scenarios by integrating the real time external API data such as traffic conditions, travel time, and opening hours. • This proposed approach will eliminate the need of custom trained machine learning datasets and make scalable, cost-effective and suitable for regions with limited structured tourism data.
KUMARASINGHE K.G.A.M.S.D.C IT22224866	Develop and deploy an AI-assisted community feedback and travel nudging system that uses real	• Develop a Community Feedback Engine that merging tourist reviews (text, voice, photo, video) with service data from	This component introduce a community-driven travel feedback and nudging system that replace static star ratings with a dynamic,

	tourist experiences to evaluate service quality and safety, and guide future travelers with reliable recommendations through intelligent analysis of customer feedback.	hotels and vehicles to create a unified trust data stream. • Implement LLM-driven Semantic Analysis using Gemini for interpret the sentiment and safety of feedback (like "Hotel was dirty") to automatically identify honest and fake reviews without human moderation. • Engineer a Trust Scoring System that assign reliability scores to services. It cross-check multiple user feedback to prevent fake or exaggerated ratings. • Create a Cognitive Nudging Interface that visualize trust scores and AI generated suggestions (e.g., "Better hotel nearby") base on user preferences. • Test and validate the system in sample scenarios with fake reviews or safety concerns to check accuracy of AI analysis and nudging effectiveness.	trust-based recommendation ecosystem. Unlike traditional tourism platforms which mostly rely on popularity, this research uniquely fuse LLM-based semantic analysis with real tourist experiences. It provide a system where every tourist act as a sensor, submitting honest feedback verified by AI against multiple user reports. This ensure that service ratings and suggestions are context-aware. This approach address the common problem of fake reviews in Sri Lanka tourism sector and guide travelers to safer services.

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10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	X	No	

b) The proposed topic exhibits novelty.

Yes	X	No	

c) The student group can successfully execute the proposed project.

Yes	X	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	X	No	

e) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor		Dilani	Lunugalage	
Co-Supervisor		Dulani	Jayasinghe	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				